University of Glasgow

ENG4042

Control 4

James Watt School of Engineering

IMPORTANT STUDENT INSTRUCTIONS

Ensure your **Student Number** and **date of birth** are **entered** in the appropriate fields on **EVERY** sheet you use, including pink continuation sheets.

Use **BLACK INK** only. DO NOT USE BLUE INK OR PENCIL.

MUST BE COMPLETED

Student number (7 numbers): 2 8 3 9 7 6 8 (N N N N N N N)

Date of Birth (DD/MM/YYYY): 09 / 03 / 2004

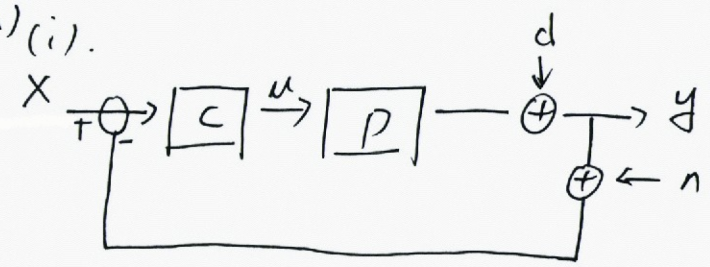
Mark with an **[X]** which question you have completed on this sheet. Cross **one box only**.

☐ DO NOT USE ☐ DO NOT USE ☒ Q3 ☐ Q4 ☐ Q5 ☐ Q6 ☐ DO NOT USE ☐ DO NOT USE

You may use both sides but only answer **ONE** question per sheet

Q3.

(a)(i).



$\Rightarrow (y+n)\cdot CP + d = y$

$\Rightarrow \frac{y(t)}{x(t)} = \frac{CP}{1+CP}x + \frac{1}{1+CP}d - \frac{CP}{1+CP}n$ ✓

$= T_o x + S_o d - T_o n$

in which $\begin{cases} S_o = \frac{1}{1+CP} \\ T_o = \frac{CP}{1+CP} \end{cases}$ ✓

Peaking in So and To should be avoided because

- it will reduce the stability robustness.
- make the response more oscillatory. ?

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(ii) The vector margin : $S_m = \min|1+CP_o|$

$\Rightarrow$ The complementary vector margin $r_m = \min\left|1+\frac{1}{CP}\right|$ ✓

$= \frac{1}{\max\left|\frac{CP}{1+CP}\right|} = \frac{1}{\max|T_o|}$

Also : $S_m = \min|1+CP| = \frac{1}{\max\left|\frac{1}{1+CP}\right|} = \frac{1}{\max|S_o|}$ ✓

The peaking of $|S_o|$ and $|T_o|$ should be avoided because it will lead to small value of vector margin. This means in the nyquist plot, the pole is very near to -1, and therefore very near to the imaginary axis on the s-plane. From $s = \delta + j\omega_n = \zeta\omega_n + j\omega_n$ ✓ we can know its damping rate is very small at this time. Therefore, it is strongly oscillatory.

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(iii). $S_{P_o}^{T_o} = \frac{P_o}{T_o}\frac{\partial T_o}{\partial P_o} = \frac{P_o}{\frac{C_oP_o}{1+P_oC_o}}\frac{\partial}{\partial P_o}\left(\frac{C_oP_o}{1+P_oC_o}\right) = \frac{1}{1+P_oC_o}$ $= S_o$

$\Rightarrow S_{P_o}^{S_o} = \frac{P_oC_o}{1+P_oC_o}$ $= -T_o$

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TOTAL 1 6 ✓

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Q Continued

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(b).
The loop-shaping design method can be used to design controller using the open-loop gain $|L_0|$

principles : ~~designing~~ appropriate $|S_0|$ and $|T_0|$, ~~the system~~.

objectives : * avoid peaking of $|S_0|$ and $|T_0|$ in the crossover frequency of $|L_0|$

* ensure $|S_0|$ small at low frequency.
* ensure $|T_0|$ small at high frequency.

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(c) The pure proportional controller can only change the magnitude of the frequency response but not the phase. It can be used to increase the bandwidth of a system by increasing its crossover frequency. However, it can also lead to reduced phase margin.

Limitation:

① Non-zero steady state error. Because there is no pure integrator, the e_{ss} may not be zero.

② Plant capacity. The control signal may overpass the plant capacity due to the P-controller.

③ Reduced phase marge (Stability robustness).